THERMAL TEST INTELLIGENCE CENTER

Engineer User Guide

Windows Portable Pilot | Application baseline: Stage 2D.4.1

A practical guide to loading thermal logger data, exploring channels, reviewing engineering indicators, comparing test runs, and exporting reports.

Before you begin

- TTIC is a local browser-based engineering analysis tool. The portable pilot is launched using TTIC.exe; a separate Python installation is not needed on the engineer's computer.
- This pilot is for engineering evaluation. Automated stability and response findings are screening indicators, not certified device pass/fail conclusions.
- Use approved, nonconfidential data until company IT authorizes the software and handling of engineering files.

Quick start — five steps

1. Extract the entire supplied TTIC folder to a local drive. Keep TTIC.exe and the _internal folder together.
2. Double-click TTIC.exe. Allow a short startup period; the browser should open at http://127.0.0.1:8501.
3. Upload a logger CSV, XLSX or XLSM file using the upload control at the top.
4. For Excel, choose the worksheet; confirm the timestamp column and check the observation, channel and duration indicators.
5. Use the six tabs to investigate results. Close the TTIC console window when finished.

Navigation at a glance

Tab	Use it for
① **Test Setup**	Check data quality; document test identity, setup and engineer review.
② **Visualization**	Explore channels on interactive dual-axis charts and export chart settings.
③ **Engineering Analytics**	Review descriptive statistics and other channel-level metrics.
④ **Event Intelligence**	Screen stability and measurement events; export detected records.
⑤ **Reports & Export**	Create individual-test Excel/PDF reports and export selected/processed data.
⑥ **Test Comparison**	Compare Test A and B; evaluate rates, stability and command response; export comparison reports.

1 Load and verify a test file

6. At the top of TTIC, select Upload a test file. Accepted types are CSV, XLSX and XLSM.
7. If using Excel, select the correct worksheet. Confirm the Timestamp column. TTIC derives elapsed seconds internally without replacing the original timestamp column.

8. Review Observations, Numeric channels, Duration and Median sample interval. Open ① Test Setup for timestamp-quality indicators and measurement groups.
9. Check warnings for backward or duplicate timestamps. If readings or channels are unexpectedly missing, review the source file and timestamp selection before proceeding.

Important: A numeric column must contain sufficiently many usable numeric values to be recognized as a measurement channel. The tool infers channel groups from names; it does not independently verify sensor units or physical meaning.

2 Document the test

10. In ① Test Setup, enter available Test ID, date, engineer, device/model, serial number, hardware revision and test type.
11. Record objective and operating conditions. Add engineer observations, conclusion and follow-up actions only after reviewing the evidence.
12. Choose Draft, Under Review or Approved as an administrative review status; this is not an electronic signature or independent technical approval.
13. If you change documentation after preparing a report, prepare the report again so the export captures the updated notes.

3 Explore the measurements

14. Open ② Visualization. Optionally filter measurement groups and select primary-axis and secondary-axis channels.
15. Use chart controls to inspect traces, zoom into a time period and compare channel behavior. Choose sensible axes and verify measurement units.
16. Use chart smoothing only to improve display readability. Engineering calculations are based on raw measurement values, not the smoothed plot.
17. Download the interactive chart as HTML or save chart configuration as JSON for reuse. A saved configuration can be loaded in a later session.

4 Review analytics and events

18. Open ③ Engineering Analytics to review descriptive metrics and export available engineering metrics as CSV.
19. Open ④ Event Intelligence to review stability screening and detected events. Export stability windows, grouped incidents and detected events where offered.

Interpretation rule: A low-variation window is a screening result, not proof of thermal equilibrium. Confirm sampling quality, channel units, test phase and physical context before drawing conclusions.

5 Compare two test runs

20. First load a primary file at the top of TTIC, then open ⑥ Test Comparison.
21. Upload Test A and Test B independently. For each Excel file, select its worksheet and timestamp column.
22. Select the measurement channel for each test. Confirm the channels represent the same physical quantity with compatible units.
23. Choose a comparison mode: Full Test (entire recordings), Common Duration (same elapsed-time extent), or Independent Windows (different start/end windows).
24. Inspect the overlay and summary statistics. Differences are descriptive: they do not automatically prove one design is better.

Comparison modes

Mode	When to use
Full Test	Overall review when both complete recordings are relevant; durations may differ.
Common Duration	Restrict both tests to the same elapsed-time span; this does not guarantee equivalent phases.
Independent Windows	Compare manually selected phases with different original start/end times.

6 Compare engineering behavior

Heating and cooling rates

Review endpoint change, overall rate and maximum positive/negative interval rates. Confirm the selected interval length and interpret rates in recorded measurement units per minute. Short, noisy intervals may produce large extrema.

Steady-state screening

Review eligible windows, qualifying low-variation windows and grouped periods. Screening uses configurable tolerance, window duration, reading count and time coverage. A grouped period may span a wider range than the individual qualifying windows.

Command-to-response analysis

Select command and response channels for each test. Use detected command steps or a manual event time. Review baseline, final-reference estimate, response direction, t10, t50, t90 and 10–90% transition time. Confirm event detection settings, signal quality and whether the selected event is comparable between tests.

A shorter measured response time describes the selected event only; it is not a universal performance ranking without equivalent test conditions.

7 Export and review reports

25. Use ⑤ Reports & Export for individual-test reporting and data exports. Use ⑥ Test Comparison for comparison Excel/PDF reports.
26. Confirm channel selection, time windows, engineering settings, test metadata and reviewer notes before generating reports.

27. Open the Excel Executive Summary and check the embedded comparison chart. Inspect detailed sheets for statistics, rates, stability and response metrics.
28. Open the PDF and confirm the response-time table includes t10, t50, t90 and 10–90% time when response results are available.
29. Save the reports with a descriptive test identifier. Retain the original logger files separately; exports do not replace source data.

Pilot test checklist

- Launch: TTIC starts from TTIC.exe without separately installed Python.
- Import: A representative CSV and Excel workbook load; worksheet and timestamp selection work.
- Data quality: observation counts, channel counts, duration and timestamp warnings match expectations.
- Visualization: channel selection, axes, zoom, HTML export and chart configuration work.
- Analytics: spot-check statistics and engineering calculations independently against raw logger readings.
- Comparison: test all three time-window modes using representative Test A/Test B files.
- Events: inspect detected steps and verify selected response metrics by hand for at least one event.
- Reporting: Excel/PDF open correctly; values and review notes agree with on-screen selections.
- Usability: record confusing labels, slow operations, errors and suggested improvements.

Record each result in ENGINEER_TEST_CHECKLIST.csv, supplied with the pilot build kit. Include the exact error text and reproduction steps; do not share sensitive logger data without approval.

9 Troubleshooting

Issue	What to try
Nothing opens	Wait briefly; check the console for errors. Confirm the full folder, including _internal, was copied.
Browser cannot connect	Check that TTIC.exe is still running. Try http://127.0.0.1:8501; another process may be using port 8501.
Windows blocks the program	Ask IT to review or allow the executable; do not bypass corporate controls.
File fails to load	Confirm file type, worksheet, valid timestamp column and readable numeric data.
Unexpected channel count	Check source columns for blanks, text, unit labels or missing readings.
Event not detected	Check whether the command channel actually changes; review threshold and event settings or use manual event time.
Report does not reflect edits	Regenerate the report after changing test documentation or settings.

10 Security and engineering limitations

- TTIC binds its local server to 127.0.0.1 in the supplied Windows pilot launcher. This is not a hosted cloud service.

- Do not assume the application or the Windows computer is certified for sensitive information. Obtain IT approval and follow company data-handling policies.
- The application does not validate calibration, sensor placement, channel units, test equivalence or engineering acceptance limits automatically.
- Rates, stable periods and command response times require human interpretation; do not treat automated screening as a safety-critical approval.
- Keep original files and independently verify any results used in formal engineering decisions.

Pilot feedback and sign-off

Engineer: _____ Date: _____

Device / test files: _____

Overall result: ☐ Works as expected ☐ Works with issues ☐ Blocked

Top issues / suggested improvements: _____

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This guide describes the Stage 2D.4.1 engineering pilot, not a validated production release.